

Cluster Number-Count Forecast with Mass–Observable Scatter: Method Summary

Xin Tang

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1 Model

1.1 Halo mass function

The comoving number density of haloes per unit halo mass is dn/dM , computed from the Tinker et al. (2008) fitting function via the `hmf` package. The expected number of haloes with true mass in bin $[M_j, M_{j+1}]$ and redshift in $[z_{\min}, z_{\max}]$ over a survey of area Ω_s is

$$N_j^{\text{true}}(\Omega_m, \sigma_8) = V_{\text{shell}}(\Omega_m, \Omega_s) \int_{M_j}^{M_{j+1}} \frac{dn}{dM} dM, \quad (1)$$

with the comoving volume of the shell

$$V_{\text{shell}} = \frac{\Omega_s}{\Omega_{\text{sky}}} [V_c(z_{\max}) - V_c(z_{\min})], \quad \Omega_{\text{sky}} = 41253 \text{ deg}^2. \quad (2)$$

1.2 Mass–observable relation

We adopt the standard log-normal MOR (Kaiser 1986; Stanek et al. 2006; Evrard et al. 2014),

$$\ln O = \ln A + \alpha \ln \left(\frac{M}{M_{\text{piv}}} \right) + \beta \ln \left(\frac{1+z}{1+z_{\text{piv}}} \right) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_{\ln O}^2). \quad (3)$$

Defining the *inferred mass* $\hat{M}$ by inversion of the mean relation and subtracting from Eq. (??), the normalisation, pivot, and redshift terms cancel, leaving

$$\ln \hat{M} = \ln M + \frac{\varepsilon}{\alpha} \implies P(\ln \hat{M} | \ln M) = \mathcal{N}(\ln M, (\sigma_{\ln O}/\alpha)^2). \quad (4)$$

Converting to base-10 logarithms,

$$\boxed{\sigma_{\log_{10} M} = \frac{\sigma_{\ln O}}{\alpha \ln 10} \approx \frac{f_{\text{obs}}}{\alpha \ln 10}}, \quad (5)$$

where $f_{\text{obs}} \approx \sigma_O/\langle O \rangle$ is the fractional scatter in the observable (valid for $\sigma_{\ln O} \lesssim 0.3$). Because Eq. (??) contains only the ratio $\sigma_{\ln O}/\alpha$, the forecast is observable-agnostic: A , M_{piv} , β , and z_{piv} do not enter.

1.3 Migration matrix

On a grid of $\log_{10} M$ bin edges $\{x_i\}$ with true-bin centres $\bar{x}_j$, the probability that a true-bin- j halo is placed in observed bin i is

$$P_{ij} = \Phi \left(\frac{x_{i+1} - \bar{x}_j}{\sigma_{\log_{10} M}} \right) - \Phi \left(\frac{x_i - \bar{x}_j}{\sigma_{\log_{10} M}} \right), \quad \sum_i P_{ij} \equiv 1, \quad (6)$$

with Φ the standard-normal CDF. Two matrices are stored: $\mathbf{P}_{\text{data}}$ built from the true (data-generating) scatter, and $\mathbf{P}_{\text{model}}$ built from the scatter assumed in the fit. The two need not be equal.

1.4 Observed counts and likelihood

The expected observed counts per bin are

$$N_i^{\text{obs}}(\Omega_m, \sigma_8) = \sum_j P_{ij} N_j^{\text{true}}(\Omega_m, \sigma_8). \quad (7)$$

Adopting a Poisson likelihood (independent bins, no shot-noise covariance),

$$\ln \mathcal{L}(\Omega_m, \sigma_8) = \sum_i \left[N_i^{\text{data}} \ln N_i^{\text{obs}} - N_i^{\text{obs}} - \ln \Gamma(N_i^{\text{data}} + 1) \right]. \quad (8)$$

The mock data vector N_i^{data} is generated once at fiducial $(\Omega_m^{\text{fid}}, \sigma_8^{\text{fid}})$ using $\mathbf{P}_{\text{data}}$; no Poisson noise is added, so the recovered posterior is centred on the fiducial when the model is correctly specified.

2 Forecast scenarios

Four assumption modes are supported by varying the model-side scatter $f_{\text{obs}}^{\text{model}}$ relative to the data-side $f_{\text{obs}}^{\text{data}}$:

Scenario	Setting	Question addressed
Correct assumption	$f^{\text{model}} = f^{\text{data}}$, fixed	Best-case statistical precision
Ignore scatter	$f^{\text{model}} = 0$	Bias from neglecting scatter
Wrong scatter value	$f^{\text{model}} \neq f^{\text{data}}$, fixed	Sensitivity to miscalibration
Marginalise over scatter	f^{model} sampled with prior	Realistic constraints with unknown scatter

3 Input parameters

3.1 Fixed configuration

Symbol	Value	Description
h	0.7	Reduced Hubble constant
Ω_m^{fid}	0.318	Fiducial matter density
σ_8^{fid}	0.80	Fiducial power-spectrum amplitude
$z_{\text{min}}, z_{\text{max}}$	0.3, 0.5	Redshift shell
HMF model	Tinker08	From <code>hmf</code> package
Number of mass bins	10	Linear in $\log_{10} M$
α_{MOR}	1.0	Default MOR mass slope
Sampler	<code>emcee</code> , 40 walkers	CosmoSIS backend

3.2 Forecast grid

Axis	Values	Notes
Mass threshold M_{min}	$5 \times 10^{13}, 2 \times 10^{14} M_{\odot} h^{-1}$	$M_{\text{max}} = 5 \times 10^{15} M_{\odot} h^{-1}$
Survey area Ω_s	1000, 2000, 4000, 8000 deg ²	X-ray-like survey
Fractional scatter f_{obs}	0.05, 0.10, 0.20	In the observable
Assumption mode	correct / ignore / wrong / marginalise	See table above

3.3 Sampled parameters and priors

Parameter	Prior	Sampled when
Ω_m	Uniform(0.1, 0.5)	Always
σ_8	Uniform(0.6, 1.0)	Always
$f_{\text{obs}}^{\text{model}}$	Uniform(0.0, 0.4)	Marginalise-over-scatter mode

4 Outputs

For each grid point the pipeline produces an MCMC chain in (Ω_m, σ_8) (and $f_{\text{obs}}^{\text{model}}$ when marginalising). The headline forecast products are:

1. 68% and 95% credible contours in the (Ω_m, σ_8) plane for each scenario.
2. Bias of the posterior mean from the fiducial point, as a function of the mismatch $f^{\text{data}} - f^{\text{model}}$.
3. Constraint area (or figure of merit) versus survey area Ω_s and mass threshold M_{min} .

5 References

- Kaiser, N. 1986, MNRAS 222, 323 — self-similar cluster scaling relations.
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